

Q5. CONCEPT MAP: COMPARING HIERARCHICAL, NETWORK, AND RELATIONAL DATA MODELS

DATA MODELS Ways to organize and represent data in a database

HIERARCHICAL DATA MODEL

Organizes data in a tree-like
structure (Parent → Child)

One-to
-Many

One
parent

Highly
organiz
ed

Less
flexible



Organization
Company → Dept → Employee

NETWORK DATA MODEL

Organizes data as a graph of
records

Many-to-
Many

Multiple
parents

Flexible

Complex



Supply Chain
Supplier → Product → Customer

RELATIONAL DATA MODEL

Organizes data in tables
(rows & columns)

Tables

Primary/
Foreign
Keys

Easy to
use

SQL &
Normali
zation



Banking
Customer → Account → Transact
ion

COMPARISON SUMMARY

Structure: Tree | Graph | Table

Flexibility: Hierarchical < Network < Relational

Data Independence: Relational is highest

Ease of Use: Relational is most user-friendly

Q10. CONCEPT MAP: ROLES AND RESPONSIBILITIES IN A DBMS ENVIRONMENT

DBMS ENVIRONMENT People and roles involved in managing and using a database

DBA (Database Administrator)

Overall control and
management

Define
e
sche
ma

Mana
ge
users

Data
securi
ty

Backu
p &
recov
ery

Perfo
man
ce
tunin
g

Goal:
Secure & efficient DB

END USER (Users of the System)

Interact using
applications/queries

Naive
User

Casual
User

Sophistic
ated
User

Follow policies
and use data

APPLICATION DEVELOPER (Programmer / Developer)

Build applications
for database

UI
Desig
n

Codin
g

Valid
ation

Maint
ain

Goal:
Reliable apps

COMMON OBJECTIVE

Ensure accurate, secure, consistent and efficient management and use of
data.